

3.3 Trigonometry

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Use addition formulae, double-angle identities and $R \cos(x-\alpha)$ transformations. A transformed form reveals amplitude and phase; choose α using both sine and cosine coefficients. For equations, list every solution over the specified interval and reject extraneous values from squaring.

Worked example: follow the reasoning

$3\sin x + 4\cos x = 5\sin(x+\alpha)$ where $\cos\alpha=3/5$ and $\sin\alpha=4/5$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Write $5\cos x + 12\sin x$ as $R\cos(x-\alpha)$ with $R>0$ and α in first quadrant.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$R = \sqrt{25 + 144} = 13$ [1]; $R \cos \alpha = 5$, $R \sin \alpha = 12$ [1]; expression $13 \cos(x - \arctan(12/5))$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.